

# DAY 18: DATA ANALYTICS MASTER CLASS

## Module: One-Sample T-Test (Comprehensive Bilingual Master Guide)



### 1. THE CORE PHILOSOPHY & BACKGROUND (बुनियादी सिद्धांत)

**English:** In real-world manufacturing, jewelry accounting, and data analytics, we rarely have access to the entire history of a population consisting of millions of data points.

*Hindi: Real-world business, jewelry manufacturing, aur accounting me hamare paas hamesha saalon ka ya lakhon records ka bada data (Population) nahi hota.*

**English:** When your Sample Size is small ( $N < 30$ ) and the population standard deviation ( $\sigma$ ) is unknown, the traditional Z-Test fails completely.

*Hindi: Jab hamare paas Sample Size chota ho ( $N < 30$ ) aur hume poori population ki asli Standard Deviation ( $\sigma$ ) na pata ho, tab Z-Test poori tarah fail ho jata hai.*

**English:** This is exactly where the Student's T-Test comes to the rescue to handle small sample variations safely.

*Hindi: Aise critical situations me chote samples ke variations ko safely handle karne ke liye kaam aata hai Student's T-Test.*

**English:** The core objective of this test is to verify whether the difference between a small sample's average and a fixed target standard is a real, significant change or just a temporary random fluctuation caused by pure luck.

*Hindi: Is test ka main maqsad yeh check karna hai ki hamare ek chote sample ka average ( $\bar{X}$ ), hamare fixed target standard ( $\mu$ ) se jo alag aa raha hai, kya woh badlaaw sach me koi asli badlaaw hai ya bas kismat ka ek chota-mota ittefaq (By Chance deviation) hai.*



### 2. CRITICAL CONDITIONS FOR USING ONE-SAMPLE T-TEST (कब aur kahan use karein?)

**English:** You cannot apply the One-Sample T-Test everywhere blindly; certain mathematical assumptions must hold true.

*Hindi: Aap har jagah aankh band karke One-Sample T-Test nahi laga sakte; isko chalane ke liye kuch mathematical conditions ka pura hona zaroori hai.*

**English:** To run this statistical engine properly, the following 4 main conditions must be satisfied completely:

Hindi: Is statistical engine ko sahi se chalane ke liye niche di gayi 4 main conditions ka pura hona ekdum zaroori hai:

1. **English: One Continuous Variable:** Your target data must be in numerical form that can be measured continuously, such as weight, length, gold purity percentage, or production units. It cannot be categorical data like Yes/No or Pass/Fail.

Hindi: Ek Naapne Wali Value: Aapka data numbers me hona chahiye jise naapa ja sake (jaise weight in grams, length in mm, pure gold percentage, ya daily production count). It should not be categorical (like Yes/No, Pass/Fail).

2. **English: Known Standard Target ( $\mu$ ):** You must have a pre-defined historical average, a government benchmark, or a company baseline to compare your sample against.

Hindi: Ek Pehle Se Tay Standard Target: Aapke paas compare karne ke liye ek pehle se tay shuda standard target, government benchmark, corporate SLA, ya purana history average hona zaroori hai.

3. **English: Independent Observations:** Each data point in your sample must be separate and independent. The weight of one gold coin should not influence or depend on the weight of another coin.

Hindi: Azaad Data Points: Ek coin ka weight dusre coin ke weight par depend nahi karna chahiye. Har ek sample azaad hona chahiye (No correlation between rows).

4. **English: Normal Distribution:** Because the sample size is small ( $N < 30$ ), the data distribution must approximately follow a bell curve, meaning it should be free from extreme anomalies or wild outliers.

Hindi: Normal Distribution (Bell Curve): Kyunki sample chota hai ( $N < 30$ ), isliye data ka distribution lagbhag normal hona chahiye, yaani usme koi ajeeb extreme values (Outliers) nahi honi chahiye jo poore average ko hila dein.



## 3. REAL-WORLD INDUSTRIAL APPLICATIONS (Kahan-Kahan Use Hota Hai?)

### A. Jewelry Manufacturing & Gold Smithing (Jewelry Industry Me Use)

- **English: Gold Purity Check:** Government regulations mandate that 22K jewelry must maintain a strict purity level of 91.6% pure gold. As an analyst, you take a random sample of 15 ornaments from a new artisan batch to ensure the average purity meets this legal standard.

Hindi: Government ka rule hai ki 22K jewelry me 91.6% pure gold hona chahiye. Ek analyst ke roop me, aapne artisans ke naye batch se bane 15 ornaments ka sample liya aur check kiya ki unka average purity 91.6% ke barabar baith raha hai ya machine/labor sona kam daal rahi hai.

- **English: Weight Precision Counter:** If a high-end luxury collection specifies that all luxury diamond rings must have a standard baseline weight of exactly 5.0 grams, you randomly pull 12 rings to

verify if the machine scaling is perfectly aligned.

*Hindi: Agar kisi special collection ke rings ka standard weight 5.0 grams fix kiya gaya hai, to randomly 12 rings uthakar unke average weight ko target se match karna.*

## B. E-Commerce & Logistics (Shipping & Delivery)

- **English: Delivery Time Benchmark:** A major logistics corporation sets an internal benchmark stating that standard shipments must be delivered within 48 hours. Sampling 20 recent shipments from the Gorakhpur hub reveals whether the hub is maintaining efficiency or causing delays.

*Hindi: Ek courier company ka target hai ki har shehar me delivery 48 ghante me ho jani chahiye.*

*Gorakhpur hub ke pichle 20 orders ka sample uthakar check karna ki kya sach me average delivery time 48 hours hai ya delivery delay ho rahi hai.*

## C. Finance & Business Accounting (Auditing)

- **English: Invoice Error Verification:** If a historical corporate audit standard claims that processed invoices have a maximum allowable average error margin of \$0.02, testing a sample of 15 recent bills confirms if the current accounting team is working within safe margins.

*Hindi: Agar company ka historical standard hai ki invoices me average error margin 0.02 se kam hona chahiye, to naye accountant ke banaye 15 bills ka sample lekar check karna ki error standard limit ke andar hai ya nahi.*

## 4. THE LIVE BUSINESS CASE (Manual Numerical Breakdown)

- **Standard Target ( $\mu$ ):** 10.0 grams per gold coin.
- **Sample Size (N):** 10 Coins
- **Asli Production Data (X):** [9.8, 10.1, 9.9, 10.2, 9.7, 10.1, 9.9, 10.0, 9.8, 10.1]

The Master Formula:

$$t = (\bar{X} - \mu) / (s / \sqrt{N})$$

### Step-by-Step Manual Solution:

#### Step A: Calculate Sample Mean ( $\bar{X}$ )

$\bar{X} = (9.8 + 10.1 + 9.9 + 10.2 + 9.7 + 10.1 + 9.9 + 10.0 + 9.8 + 10.1) / 10 = 99.6 / 10 = 9.96$  grams

**\*(Hamari machine ka average target se -0.04 grams peeche chal raha hai).\***

#### Step B: Calculate Sample Standard Deviation (s)

Formula:  $s = \sqrt{[\sum(X - \bar{X})^2 / (N - 1)]}$

Squares of individual deviations from the mean:

- $(9.8 - 9.96)^2 = 0.0256$  | •  $(10.1 - 9.96)^2 = 0.0196$  | •  $(9.9 - 9.96)^2 = 0.0036$
- $(10.2 - 9.96)^2 = 0.0576$  | •  $(9.7 - 9.96)^2 = 0.0676$  | •  $(10.1 - 9.96)^2 = 0.0196$
- $(9.9 - 9.96)^2 = 0.0036$  | •  $(10.0 - 9.96)^2 = 0.0016$  | •  $(9.8 - 9.96)^2 = 0.0256$

- $(10.1 - 9.96)^2 = 0.0196$

Sum of squared differences:  $\sum(X - \bar{X})^2 = 0.244$

$$s = \sqrt{[0.244 / (10 - 1)]} = \sqrt{0.02711} = \mathbf{0.1646 \text{ grams}}$$

#### Step C: Calculate Standard Error (S.E.)

$$S.E. = s / \sqrt{N} = 0.1646 / \sqrt{10} = 0.1646 / 3.1622 = \mathbf{0.05205 \text{ grams}}$$

#### Step D: Calculate Final T-Score (The Multiples Check)

Mean Difference =  $9.96 - 10.0 = -0.04 \text{ grams}$

$$t = -0.04 / 0.05205 = \mathbf{-0.7685}$$

## 5. THE GRAND MASTER OBSERVATION (The "Guna" Framework)

**English Intuition:** Standard Error (S.E.) represents the fundamental baseline unit of luck or natural fluctuation inherent in small data sampling.

*Hindi: Standard Error (S.E.) statistics ki duniya me ek Unit (Scale ya Paimana) ki tarah kaam karta hai jo batata hai ki kismat ke bharose ek normal natural jhatka kitna bada ho sakta hai.*

**English:** By dividing the calculated Mean Difference by the Standard Error, we are mathematically discovering exactly how many multiples (Guna) of natural fluctuation our actual difference represents.

*Hindi: Jab hum Mean Difference ko S.E. se divide karte hain, to hume yeh pata chalta hai ki hamara asli difference kismat ke jhatke se kitna GUNA (multiples) bada ya chhota hai!*

**English Study Case:** For instance, if a dataset reveals a Mean Difference of 4 and a Standard Error of 2, the division yields a T-score of 2. This proves that the actual difference is generating a shockwave that is exactly 2 steps or 2 multiples larger than a completely normal variance.

*Hindi Example: Agar kisi data me Mean Difference = 4 hai aur uski S.E. = 2 hai, to  $4 / 2 = 2$ . Iska matlab jo difference aaya hai, woh S.E. se 2 guna / 2 kadam zyada jhatka de raha hai.*

## 6. FINAL INDUSTRIAL VERDICT & INTERPRETATION (Industrial Faisla)

**English:** Our calculated final T-score stands at -0.7685 steps away from the baseline center.

*Hindi: Hamara final calculated T-Score baseline center se -0.7685 kadam door aaya hai.*

**English:** Because our actual mathematical difference is significantly smaller than even 1 full multiple of standard error, it falls well within the territory of acceptable natural variation.

*Hindi: Kyunki hamara final farq kismat ke 1 guna jhatke se bhi chhota hai (sirf -0.76 kadam door hai), iska matlab yeh difference bilkul normal aur acceptable natural variation ke andar aata hai.*

**English:** Minor human hand variations or melting temperature cooling cycles can easily cause this tiny deviation without indicating any underlying machine malfunction.

*Hindi: Labor ke hath hilne se ya gold melting process me cooling cycles ki wajah se itna halka variation aana bilkul normal aur safe baat hai.*

**English:** Therefore, the Null Hypothesis ( $H_0$ ) stands victorious, indicating that the machinery is perfectly aligned and requires absolutely zero mechanical tuning or structural maintenance.

*Hindi: Isiliye, Null Hypothesis ( $H_0$ ) jeet gaya, jiska matlab hai ki machinery ekdum perfect kaam kar rahi hai aur isme kisi bhi tuning ya structural maintenance ki koi zaroorat nahi hai. Data ekdum clear hai!*